Cross-Modal Action Recognition in Egocentric Video Using Mamba: Integrating RGB and Hand Skeleton Streams via CLS Token Fusion Strategies

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Abstract

Egocentric action recognition is a challenging task due to erratic camera motion, frequent hand occlusion, and the difficulty of maintaining consistent visual representations over time. In this work, we propose a cross-modal architecture that combines RGB video and temporal hand skeleton data within a unified Mamba-based framework, exploiting the linear time complexity of State Space Models (SSMs). Our architecture consists of three components: a VideoMamba module for visual feature extraction, a skeleton encoder built on a stack of Mamba blocks, and a fusion module that integrates both modalities into a single representation. A central contribution of this work is the design and evaluation of four Class (CLS) token mixing strategies for multimodal fusion: Naive, Average, Weighted and Context-based. These strategies differ in how the pretrained unimodal CLS tokens, which role is to act as information sinks concentrating learned representations, are leveraged to initialize the mixed CLS token used for final classification. We evaluate all strategies on the H2O dataset. Experimental results show that the Average strategy achieves the best performance, yielding gains of over 10% Top-1 accuracy in the Tiny configuration and 25% in the Small configuration over the VideoMamba baseline. Furthermore, we observe that the fusion mechanism exhibits robustness to uninformative modalities, degrading the contributions of noisy inputs.

1. Introduction

Egocentric videos, recorded from a first-person point of view, face several challenges that fixed cameras typically avoid, such as erratic camera motion, frequent occlusion by the hands, and the possibility of temporarily losing visual contact with the main action. For this reason, a flexible architecture capable of maintaining a global video representation while being continuously updated with noisy or misleading frames is required. These challenges make Transformers [17] particularly well suited for egocentric video understanding, as their self-attention mechanism enables the construction of strong spatio-temporal representations and allows the model to retain important visual information over long sequences, even in the presence of extended periods of misleading input.

In addition, the ability of Transformers to incorporate multimodal data into their representations allows skeletal motion information to be used to improve action prediction. Temporal hand skeleton data not only helps mitigate occlusion issues, such as when fingers are hidden behind objects or the hand itself, but also provides clean motion cues related to the performed action, thereby enhancing the overall understanding. This has become increasingly easier with the growing availability of egocentric datasets that provide synchronized hand skeleton annotations alongside the video stream [5, 13], most notably the H2O dataset [13], which captures a diverse set of everyday bimanual interactions (pouring, cutting, assembling, etc) from a first-person perspective with ground-truth hand keypoint annotations, making it particularly well suited for evaluating action recognition models in realistic daily-life scenarios.

Although these properties make Transformers a natural fit for egocentric video understanding, unfortunately the self-attention mechanism scales quadratically with sequence length, which limits the feasibility of Transformers for online video understanding applications. For this reason, Mamba [9], a State Space Model (SSM)-based architecture, has recently gained attention due to its linear time complexity and performance comparable to Transformers in NLP tasks. Extensions of Mamba to visual domains, such as Vision Mamba [20] and Video Mamba [14], have demonstrated promising results in image and video processing.

Although skeleton-based action recognition models using Mamba have been explored [4, 19], to the best of our knowledge, no prior work has integrated skeleton-based information with RGB streams within this framework. We address this gap by proposing a cross-modal Mamba architecture and systematically evaluating four strategies for fusing pretrained unimodal Class(CLS) token representations into a unified classification token.

2. Related Work

2.1. Video Understanding with Mamba

Video understanding is a fundamental challenge in computer vision, with action recognition offering a broad range of real-world applications in video surveillance, healthcare, sports analytics, autonomous driving, and robotics [10]. The growth of large-scale annotated datasets [6, 8, 12, 16] has driven significant progress by enabling richer evaluation and more robust model training. The field is broadly divided into two objects of study: short-form video understanding , where actions are contained within a few seconds, and long-form understanding , where relevant events may be sparsely distributed across minutes or hours [10]. This distinction has direct implications for architectural design, as models must balance temporal resolution with computational efficiency.

Several recent works have demonstrated the potential and advantages of adapting the vanilla Mamba block for video understanding tasks. For instance, VideoMambaSuite [2] explores replacing Transformer blocks with Mamba blocks in standard Transformer-based architectures for image and video understanding tasks, including temporal action localization, temporal action segmentation, dense video captioning, video paragraph captioning, and action anticipation. In most cases, the modified models achieved improved performance while consuming significantly less GPU memory and reducing inference time, even when compared against Transformer models using Flash Attention [3].

One of the first works to adapt Mamba blocks for video input was VideoMamba [14]. Its architecture follows a design similar to that of Video Transformers. Each frame is first divided into non-overlapping patches using a 3D convolution with a $1\times16\times16$ kernel. An additional CLS token is appended, and each patch embedding is combined with a positional embedding determined by the scanning strategy, along with a temporal embedding that encodes the frame index. The resulting sequence representation is then processed by a stack of bidirectional Mamba blocks, and the final classification is obtained from the CLS token, as illustrated in Fig. 1.

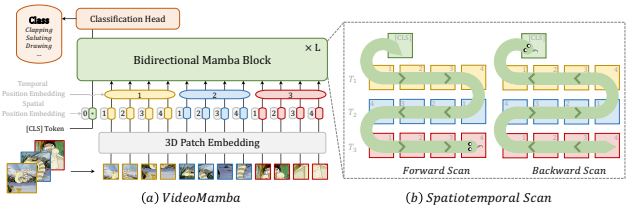

Figure 1. Schema of the inner working of a VideoMamba [14]

Their results demonstrate that the model is capable of capturing and maintaining spatio-temporal dependencies in short video datasets at performance levels comparable to or better than Transformers. Moreover, in long video sequences, VideoMamba surpasses TimeSformer [1] both in accuracy and inference time, highlighting the advantage of linear-time computation when modeling long sequences.

2.2. Skeleton-Based Action Recognition with Mamba

The use of skeleton data for action detection in video processing is not new. The concept was already explored in earlier works [11, 18] using handcrafted features. As discussed in [15], representing the human body as a moving skeleton is not only computationally efficient but also more robust to environmental variations, since it focuses exclusively on joint motion rather than surrounding context.

Due to their inherent graph structure, skeleton data is commonly modeled using Graph Convolutional Networks (GCNs), which preserve spatial relationships between joints during inference. For example, ActionMamba [19] combines GCN blocks with Mamba blocks for action recognition, leveraging Mamba's ability to capture temporal dependencies from features extracted by the GCN.

To maintain architectural simplicity, in this work we instead employ a deep stack of Mamba blocks to directly model the spatio-temporal representation of skeleton data, avoiding additional graph-based components.

3. Cross-Modal Fusion Design

3.1. Two-Branch Architecture

The overall architecture is divided into three main components: (1) a VideoMamba module to extract embeddings from the video stream, (2) a pure Mamba module to encode skeleton data, and (3) a fusion module that combines both information streams into a unified representation, as can be seen in Fig. 2.

Both the video and skeleton embeddings share the same dimensions. However, the number of tokens differs between modalities, as the video stream produces multiple tokens per frame due to spatial patchification.

The fusion module is inspired by the Mamba Suite Cross-Modal implementation [2]. They add a learnable modality embedding to each modality-specific representation before concatenating them into a single sequence. Instead of keeping the original CLS tokens from each stream, they remove both and introduce a new one. The concatenated sequence is then processed through a Mamba block, and its final value is used for classification.

Our approach differs from the original implementation in how we handle the two modality-specific CLS tokens. In Mamba models, as in Transformers, it acts as an information sink, concentrating the relevant contextual informa-

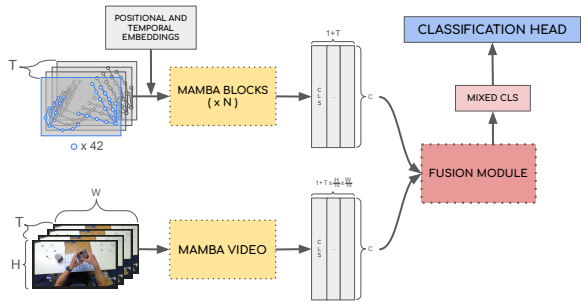

Figure 2. Overview of the proposed cross-modal architecture. The skeleton stream (top) and video stream (bottom) are encoded independently before being fused into a unified Mixed CLS token for classification. T denotes the number of input frames, H and W the height and width of each frame respectively, and C the embedding dimension.

tion throughout processing before passing it to the classification head. By using pretrained video and skeleton encoders, their respective CLS tokens already contain dense and structured information. Discarding them entirely would waste this learned behavior.

For this reason, we propose four strategies to initialize the new CLS token, which we are referring to as the *Mixed CLS token*, by leveraging the pretrained representations. A visual representation is shown in Fig. 3.

3.2. CLS Mixing Strategies

Naive. Both modality-specific CLS tokens are discarded, and a new trainable one is used.

Average. The Mixed CLS token is initialized as the average by dimension of the video and skeleton CLS tokens.

Weighted. A learnable scalar parameter ω is passed through a sigmoid function to compute a linear combination of the two CLS tokens:

$$\text{CLS}_{mix} = \alpha\,\text{CLS}_{video} + (1-\alpha)\,\text{CLS}_{skeleton}, \quad (1)$$

where $\alpha = \sigma(\omega)$.

Context-based. Instead of learning a static scalar weight, we compute ω dynamically using a Mamba block that processes the concatenated token representations (excluding both CLS tokens). Specifically:

$$\alpha = \sigma(\text{Mean}(\text{Mamba}(\mathbf{T}_{video} \oplus \mathbf{T}_{skel}))), \quad (2)$$

where $\mathbf{T}_{video}$ and $\mathbf{T}_{skel}$ denote the non-CLS token sequences from each modality, $\oplus$ denotes concatenation, and $\text{Mean}(\cdot)$ denotes taking the mean over the embedding values. The resulting α is then used in Eq. (1) to combine the two CLS tokens, allowing the model to adapt the fusion weight based on the input content.

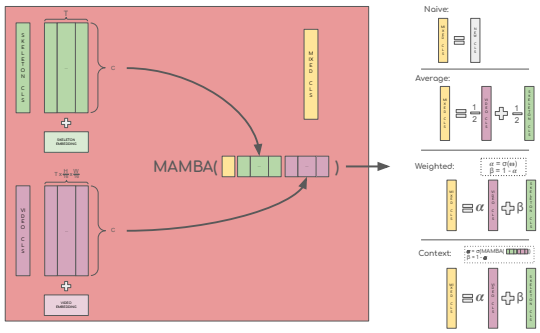

Figure 3. The four proposed CLS mixing strategies. From top to bottom: Naive (random initialization), Average (equal combination), Weighted (learnable scalar), and Context-based (input-dependent weight computed via a Mamba block). C denotes the embedding dimension.

4. Experiments and Results

4.1. Experimental Setup

To evaluate the cross-modal capabilities of Mamba and the proposed fusion strategies, we conducted experiments on the H2O dataset [13], an egocentric video dataset that provides not only action labels but also ground-truth hand keypoints for each frame.

For both the VideoMamba module and the skeleton encoder, we adopted the same configurations defined in the VideoMamba model variants [14], summarized in Tab. 1.

Table 1. Model configurations used for both the video and skeleton encoders. Depth denotes the number of stacked Mamba blocks in each branch.

Variant	Depth	Embedding Dim.
Tiny	24	198
Small	24	386
Middle	32	598

The VideoMamba visual models were initialized from publicly available checkpoints pretrained for short-term video understanding on the Something-Something V2 dataset [7], operating on 8-frame inputs. The skeleton encoder was trained directly on the H2O dataset.

We deliberately limited the fusion module to a single Mamba block, ensuring that any performance gains can be attributed to the model's ability to take advantage of the cross-modal information rather than to an increase in model depth.

Training was performed for 50 epochs using a linear warm-up schedule followed by a CosineAnnealingLR

scheduler and the AdamW optimizer. Early stopping was applied if no validation improvement was observed for 10 consecutive epochs. All 18 models were trained on a single NVIDIA RTX 5090 GPU (32 GB VRAM), with a total training time of approximately 2.5 hours.

4.2. Analysis and Results

As shown in Tab. 2, the **Average** strategy achieves the best performance improvement, obtaining a gain of more than 10% Top-1 accuracy over the VideoMamba baseline in the Tiny configuration and 25% in the Small configuration. The **Weighted** strategy follows closely for the Tiny and Small variants, suggesting that the learnable parameter converges toward a balanced combination (*i.e.*, $\alpha \approx 0.5$), effectively approximating the averaging strategy.

Table 2. Top-1 accuracy (%) on H2O for all fusion strategies compared against unimodal baselines. VideoMamba refers to the pretrained video encoder alone, and Skeleton Mamba to the skeleton-only Mamba stack. δ_V and δ_S denote absolute difference in accuracy points, while Δ_V and Δ_S denote relative change with respect to each baseline respectively.

Tiny (depth 24, dim 198)					
Method	Top-1 (%)	δ_V	Δ_V	δ_S	Δ_S
Average	61.90	+5.71	+10.17%	+35.23	+132.14%
Weighted	58.10	+1.90	+3.39%	+31.43	+117.86%
VideoMamba	**56.19**	—	—	+29.52	+110.71%
Naive	50.48	-5.71	-10.17%	+23.81	+89.29%
Context	50.48	-5.71	-10.17%	+23.81	+89.29%
Skeleton Mamba	**26.67**	-29.52	-52.54%	—	—
Small (depth 24, dim 386)					
Method	Top-1 (%)	δ_V	Δ_V	δ_S	Δ_S
Average	60.95	+12.38	+25.49%	+28.57	+88.24%
Context	60.00	+11.43	+23.53%	+27.62	+85.29%
Naive	56.19	+7.62	+15.69%	+23.81	+73.53%
Weighted	54.29	+5.72	+11.77%	+21.90	+67.65%
VideoMamba	**48.57**	—	—	+16.19	+50.00%
Skeleton Mamba	**32.38**	-16.19	-33.33%	—	—
Middle (depth 32, dim 598)					
Method	Top-1 (%)	δ_V	Δ_V	δ_S	Δ_S
VideoMamba	**58.10**	—	—	+49.53	+577.78%
Average	58.10	—	—	+49.53	+577.78%
Weighted	55.24	-2.86	-4.92%	+46.67	+544.44%
Context	54.29	-3.81	-6.56%	+45.71	+533.33%
Skeleton Mamba	**8.57**	-49.53	-85.25%	—	—
Naive	3.81	-54.29	-93.44%	-4.76	-55.56%

Inspection of the learned weight parameter α across model configurations reveals values of 0.62, 0.64, and 0.64 for the Tiny, Small, and Middle variants respectively. These values consistently fall near 0.5, confirming that the weighted model converges toward a near-equal combination of both modalities. Notably, the slight but consistent bias above 0.5 suggests that the visual stream is marginally weighted higher than the skeleton stream, which is interpretable given that the VideoMamba encoder benefits from large-scale pretraining on Something-Something V2, while the skeleton encoder is trained exclusively on the comparatively small H2O dataset.

As expected, the **Naive** strategy consistently performs worse than the other fusion approaches. Since it discards both pretrained CLS tokens and replaces them with a fresh one, it does not effectively exploit the cross-modal information learned by the unimodal encoders.

An interesting observation arises in the Middle configuration, where the skeleton-only model achieves a Top-1 accuracy of only 8.57%, indicating severe overfitting due to limited data. However, when this weak skeleton encoder is used in the cross-modal setting, the resulting performance matches that of the VideoMamba baseline. This suggests that the fusion mechanism is capable of learning to suppress noisy or uninformative modalities and rely primarily on the stronger visual stream.

Conversely, in the Small configuration, where both modalities contain meaningful information, the cross-modal model surpasses the video-only baseline even with the addition of only a single extra Mamba block. This demonstrates the complementary nature of the two modalities when both provide informative signals.

Contrary to expectations, the **Context-based** strategy does not outperform the simpler fusion methods, despite its higher flexibility. This may be due to the limited scale of the H2O dataset rather than architectural limitations.

5. Conclusions and Future Work

In this work, we introduced a cross-modal Mamba-based architecture for egocentric action recognition that fuses RGB video streams with temporal hand skeleton data. We proposed and evaluated four CLS token fusion strategies (Naive, Average, Weighted, and Context-based) designed to leverage the dense, structured information encoded in pretrained unimodal CLS tokens when initializing a joint representation for classification.

Our experiments on the H2O dataset demonstrate that combining visual and skeletal modalities within a Mamba framework consistently outperforms the video-only baseline when both modalities carry meaningful information. The Average strategy proved to be the most effective fusion approach, achieving Top-1 accuracy gains of over 10% and 25% in the Tiny and Small model configurations respectively. The Weighted strategy performed comparably, likely converging toward a balanced combination equivalent to averaging. The Naive strategy, which discards pretrained CLS tokens entirely, consistently underperformed, confirming that preserving the learned unimodal representations during fusion is critical. Notably, the Context-based strategy, despite its greater flexibility, did not surpass simpler methods, an outcome we attribute to the limited scale of the H2O dataset rather than to architectural shortcomings.

An important emergent property of the proposed fusion mechanism is its robustness to noisy or uninformative modalities. When the skeleton encoder fails to generalize,

as observed in the Middle configuration due to overfitting, the cross-modal model recovers to video-only performance levels, suggesting that the fusion module implicitly learns to suppress low-quality inputs and rely on the more informative stream. This behavior is particularly relevant for real-world deployment scenarios where data quality cannot be guaranteed.

Several lines of work have originated from these results and are currently being pursued. First, evaluating the proposed strategies on larger egocentric datasets would clarify whether the saturation observed with the Context-based strategy is a data limitation or a fundamental architectural constraint. Second, an especially compelling direction is the use of skeleton data exclusively during training as a form of privileged information or cross-modal distillation, with the goal of improving visual-only inference without requiring skeleton inputs at test time. This would make the approach practical in deployment settings where hand keypoint estimation is unavailable or computationally prohibitive.

References

[1] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. Is space-time attention all you need for video understanding?, 2021. 2

[2] Guang Chen, Yihong Huang, Jiarui Xu, Baoxiang Pei, Zongjian Chen, Zhe Li, Jian Wang, Kunchang Li, Tong Lu, and Limin Wang. Video mamba suite: State space model as a versatile alternative for video understanding, 2024. 2

[3] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness, 2022. 2

[4] Ahmed Dawood, Boris Knyazev, and Graham W. Taylor. Simba: Mamba augmented u-shiftgcn for skeletal action recognition in videos, 2024. 1

[5] Rao Fu, Dingxi Zhang, Alex Jiang, Wanjia Fu, Austin Funk, Daniel Ritchie, and Srinath Sridhar. Gigahands: A massive annotated dataset of bimanual hand activities, 2025. 1

[6] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzyńska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Müller-Freitag, et al. The "something something" video database for learning and evaluating visual common sense. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017. 2

[7] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzyńska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, Florian Hoppe, Christian Thurau, Ingo Bax, and Roland Memisevic. The "something something" video database for learning and evaluating visual common sense, 2017. 3

[8] Kristen Grauman, Andrew Westbury, Eugene Byrne, Vincent Cartillier, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, and ... Ego4d: Around the world in 3,600 hours of egocentric video. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. 2

[9] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces, 2023. 1

[10] Sehwan Heo, Junbeom Moon, and Soon Ki Jung. Action recognition: A comprehensive survey of tasks, methods, and challenges. *ICT Express*, 12(1):32–49, 2026. 2

[11] Mohamed E. Hussein, Marwan Torki, Mohamed A. Gowayyed, and Moustafa El-Saban. Human action recognition using a temporal hierarchy of covariance descriptors on 3d joint locations. In *Proceedings of the 23rd International Joint Conference on Artificial Intelligence (IJCAI)*, 2013. 2

[12] Will Kay, João Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, Mustafa Suleyman, and Andrew Zisserman. The Kinetics human action video dataset. *CoRR*, abs/1705.06950, 2017. 2

[13] Taehyun Kwon, Bugra Tekin, Jan Stühmer, Federica Bogo, and Marc Pollefeys. H2o: Two hands manipulating objects for first person interaction recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10138–10148, 2021. 1, 3

[14] Kunchang Li, Xinhao Li, Yi Wang, Yinan He, Yali Wang, Limin Wang, and Yu Qiao. Videomamba: State space model for efficient video understanding. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024. 1, 2, 3

[15] Bin Ren, Mengyuan Liu, Runwei Ding, and Hong Liu. A survey on 3d skeleton-based action recognition using learning method, 2024. 2

[16] Khurram Soomro, Amir Roshan Zamir, and Mubarak Shah. Ucf101: A dataset of 101 human actions classes from videos in the wild. *CoRR*, abs/1212.0402, 2012. 2

[17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2023. 1

[18] Raviteja Vemulapalli, Felipe Arrate, and Rama Chellappa. Human action recognition by representing 3d skeletons as points in a lie group. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 588–595, 2014. 2

[19] Jun Wen, Dong Liu, and Bin Zheng. Actionmamba: Action spatial-temporal aggregation network based on mamba and gcn for skeleton-based action recognition. *Electronics*, 14 (18):3610, 2025. 1, 2

[20] Lianghui Zhu, Bencheng Liao, Qian Zhang, Xinlong Wang, Wenyu Liu, and Xinggang Wang. Vision mamba: Efficient visual representation learning with bidirectional state space model, 2024. 1